

Experiment No: 02

Date: 15/09/2026

Title: Bank Account Management System using Inheritance and Method Overriding

Objective / Aim

To design and implement a **Bank Account Management System** in Java using **classes, objects, inheritance, constructors, methods, method overriding, deposits, withdrawals, interest calculation, and fund transfers.**

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

- Operating System: Windows / Linux / macOS
 - Java Development Kit (JDK 17 or later)
 - Text Editor or IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans
 - Command Prompt / Terminal
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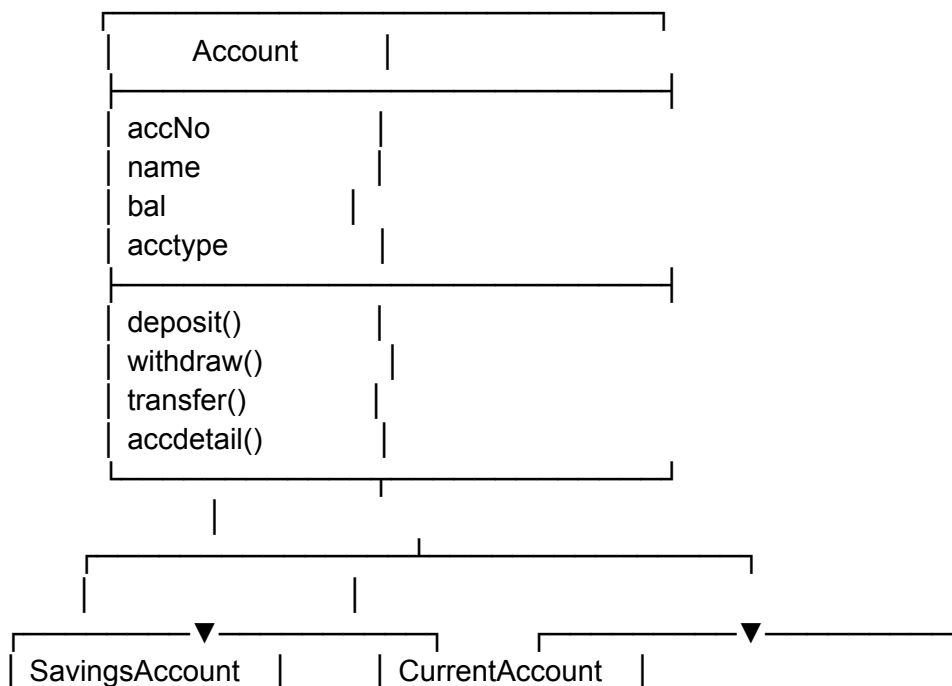
Algorithm / Logic Description

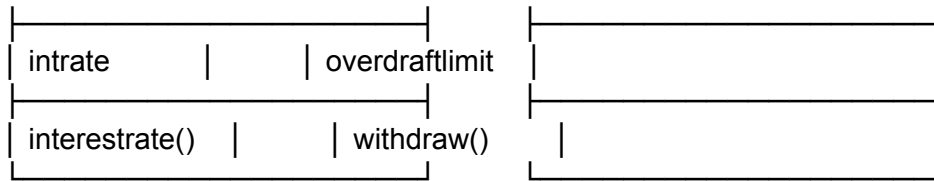
1. Create a base class named **Account**.
2. Define the following attributes:
 - Account number

- Account holder name
 - Balance
 - Account type
3. Create a constructor in the **Account** class to initialize account details.
 4. Define **deposit()** method to add money into the account.
 5. Define **withdraw()** method to withdraw money if sufficient balance is available.
 6. Define **transfer()** method to transfer money from one account to another.
 7. Define **accdetail()** method to display account holder details.
 8. Create a **SavingsAccount** class that inherits from **Account**.
 9. Add an **intrate** attribute to the **SavingsAccount** class.
 10. Define **interestrate()** method to calculate and add interest to the savings account.
 11. Create a **CurrentAccount** class that inherits from **Account**.
 12. Add an **overdraftlimit** attribute to the **CurrentAccount** class.
 13. Override the **withdraw()** method in **CurrentAccount** to allow withdrawal up to the overdraft limit.
 14. Create objects for **SavingsAccount** and **CurrentAccount**.
 15. Perform deposits, withdrawals, interest calculation, and money transfer.
 16. Display the final account details.
-



UML Class Diagram





Relationship: **SavingsAccount** and **CurrentAccount** inherit from the **Account** class.

CurrentAccount overrides the **withdraw()** method of the **Account** class.



Source Code

// creating a class for account

```
class Account {
```

```
    int accNo;
    String name;
    int bal;
    String acctype;
```

// constructor

```
Account(int accNo, String name, int bal, String acctype) {
    this.accNo = accNo;
    this.name = name;
    this.bal = bal;
    this.acctype = acctype;
}
```

// method to deposit money

```
void deposit(int amount) {
    bal = bal + amount;

    System.out.println(
        "amount is deposited into account succesfully : " + amount
    );

    System.out.println("balance updated to " + bal);
}
```

// method to remove money

```
void withdraw(int amount) {
```

```

    if (amount > bal) {
        System.out.println("balance insufficient");
    }
    else {
        bal = bal - amount;

        System.out.println(
            "withdraw suscessfully :" + amount
        );

        System.out.println(
            "balance is upated to :" + bal
        );
    }
}

```

```

// to transfer money from one account to another
void transfer(Account name, int amount) {

```

```

    if (amount > bal) {
        System.out.println("balance insufficient");
    }
    else {
        name.bal = name.bal + amount;
        bal = bal - amount;

        System.out.println("transacation done successfully");

        System.out.println(
            "balance in acc is :" + bal
        );
    }
}

```

```

// to present account holder details
void accdetail() {

```

```

    System.out.println(
        "account holder name" + name
    );

    System.out.println(
        "account no is " + accNo
    );
}

```

```

    );

    System.out.println(
        "balance in account" + bal
    );

    System.out.println(
        "account type" + acctype
    );
}
}

```

```

// creating a class for savings account
class SavingsAccount extends Account {

```

```

    double intrate;

```

```

    // constructor

```

```

    SavingsAccount(
        int accNo,
        String name,
        int bal,
        double intrate
    ) {
        super(accNo, name, bal, "savings");
        this.intrate = intrate;
    }

```

```

    // method to calculate interest rate

```

```

    void interestrate() {

        double interest = bal * intrate / 100;

        bal = bal + (int) interest;

        System.out.println(
            "interest is " + interest
        );

        System.out.println(
            "balance after interest is " + bal
        );
    }
}

```

```
}
```

```
// creating a class for current account  
class CurrentAccount extends Account {
```

```
    int overdraftlimit;
```

```
    // constructor
```

```
    CurrentAccount(  
        int accNo,  
        String name,  
        int bal,  
        int overdraftlimit  
    ) {
```

```
        super(accNo, name, bal, "current");  
        this.overdraftlimit = overdraftlimit;  
    }
```

```
    // overriding withdraw method
```

```
    @Override
```

```
    void withdraw(int amount) {
```

```
        if (amount > bal + overdraftlimit) {  
            System.out.println(  
                "withdrawal exceeds overdraft limit"  
            );  
        }
```

```
        else {  
            bal = bal - amount;
```

```
            System.out.println(  
                "withdraw suscessfully :" + amount  
            );
```

```
            System.out.println(  
                "balance is upated to :" + bal  
            );  
        }
```

```
    }  
}
```

```
// main class
```

```
public class TaskII {

    public static void main(String[] args) {

        // creating savings account object
        SavingsAccount s1 =
            new SavingsAccount(
                101,
                "Naveen",
                10000,
                5
            );

        // creating current account object
        CurrentAccount c1 =
            new CurrentAccount(
                102,
                "Rahul",
                5000,
                3000
            );

        System.out.println(
            "----- SAVINGS ACCOUNT DETAILS -----"
        );

        s1.accdetail();

        System.out.println(
            "\n----- CURRENT ACCOUNT DETAILS -----"
        );

        c1.accdetail();

        System.out.println(
            "\n----- TRANSACTIONS -----"
        );

        // depositing money into savings account
        s1.deposit(2000);

        // withdrawing money from savings account
        s1.withdraw(1000);
    }
}
```

```
// calculating interest
s1.interestrates();

// depositing money into current account
c1.deposit(2000);

// withdrawing money from current account
c1.withdraw(9000);

// transferring money from savings to current
s1.transfer(c1, 2000);

System.out.println(
    "\n----- FINAL ACCOUNT DETAILS -----"
);

System.out.println(
    "\nSavings Account"
);

s1.accountdetail();

System.out.println(
    "\nCurrent Account"
);

c1.accountdetail();
}
}
```



Compilation & Execution Commands

Save the source code exactly as:

TaskII.java

Compile the program:

```
javac TaskII.java
```

Run the program:



Output Screen

----- SAVINGS ACCOUNT DETAILS -----

account holder nameNaveen
account no is 101
balance in account10000
account typesavings

----- CURRENT ACCOUNT DETAILS -----

account holder nameRahul
account no is 102
balance in account5000
account typecurrent

----- TRANSACTIONS -----

amount is deposited into account succesfully : 2000
balance updated to 12000

withdraw suscessfully :1000
balance is upated to :11000

interest is 550.0
balance after interest is 11550

amount is deposited into account succesfully : 2000
balance updated to 7000

withdraw suscessfully :9000
balance is upated to :-2000

transacation done successfully
balance in acc is :9550

----- FINAL ACCOUNT DETAILS -----

Savings Account
account holder nameNaveen
account no is 101
balance in account9550
account typesavings

Current Account

account holder nameRahul

balance in account-2000

account typecurrent

```
PS C:\Users\navvee\OneDrive\Desktop\programs> javac TaskII.java
PS C:\Users\navvee\OneDrive\Desktop\programs> java TaskII
----- SAVINGS ACCOUNT DETAILS -----
account holder nameNaveen
account no is 101
balance in account10000
account typesavings

----- CURRENT ACCOUNT DETAILS -----
account holder nameRahul
account no is 102
balance in account5000
account typecurrent

----- TRANSACTIONS -----
amount is deposited into account succesfully : 2000
balance updated to 12000
withdraw suscessfully :1000
balance is upated to :11000
interest is 550.0
balance after interest is 11550
amount is deposited into account succesfully : 2000
balance updated to 7000
withdraw suscessfully :9000
balance is upated to :-2000
transacation done successfully
balance in acc is :9550
```

----- FINAL ACCOUNT DETAILS -----

Savings Account

account holder nameNaveen

account no is 101

balance in account9550

account typesavings

Current Account

account holder nameRahul

account no is 102

balance in account0

account typecurrent